Business Requirements Document – Role Configuration

TaskFlow – Intelligent Task Allocation & Tracking System

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> **Module:** Settings · **Sub-module:** 06.4 Role Configuration · **Code:** ROLE
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Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version
1.0	31-Oct-2026	Jojo George	Approved and signed off; baselined as the agreed business requirements.
1.0	31-Oct-2026	Jojo George	Editorial only: Related Documents paths updated to the `TF - Documents` folder layout; no requirement changed, sign-off unchanged.

Approvals / Sign-off

Role	Name	Decision	Date
Product owner	Jojo George	Approved	31-Oct-2026
Reviewer	Jojo George	Approved	31-Oct-2026

Related Documents

Document	Location
Overall BRD (roles, access model, global rules)	`../../00-Overall/BRD-TaskFlow-Overall.md`
Employee Management BRD (assigning roles to users)	`../../03-TaskFlow-Management/03.1-Employee-Management/BRD-Employee-Management.md`
Development Plan (screen-wise)	*Not applicable – not in the original 20-screen plan.*
Original BRD (section 1.4 User Roles)	`BRD for Task management.docx` (v1.0)

Document	Location
FRD	`../../../FRD/06-Settings/06.4-Role-Configuration/FRD-Role-Configuration.md`
HLD / LLD	`../../../HLD/06-Settings/06.4-Role-Configuration/HLD-Role-Configuration.md`, `../../../LLD/06-Settings/06.4-Role-Configuration/LLD-Role-Configuration.md`

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1. Project Overview

1.1 Introduction

Role Configuration is the screen where an authorized user decides exactly what each role can see and do in TaskFlow. The three system roles (Administrator, Manager, Employee) are shown read-only. The organization creates, edits, duplicates and deletes its own **custom roles** by ticking permissions from a fixed permission catalogue.

1.2 Business Problem / Current Situation

With only three fixed roles, an organization must give people either too much access (Manager) or too little (Employee). A team lead who should see workload but not manage employees, or an assistant who should only send Mail, cannot be served. Changing what a role may do would need a developer.

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1.3 Project Objectives

ID	Objective	Success measure
OBJ-ROLE-01	Let the organization define custom roles that grant exactly the screens and actions a group needs	A custom role is created, edited and deleted from the screen without code changes
OBJ-ROLE-02	Protect the built-in access model	System roles and the permission catalogue cannot be changed by anyone; Administrator-only permissions cannot be given away
OBJ-ROLE-03	Make permission changes effective at once	Every holder of a changed role is signed out and signs in again with the new access

1.4 Expected Benefits

- Least-privilege access: each group gets what it needs and nothing more.
- No risk of breaking the system roles by mistake.
- Changes are immediate and visible: a compare view shows every role against every permission.

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2. Stakeholders & User Roles

2.1 Stakeholders

Stakeholder	Interest / responsibility
Project owner (Jojo George)	Defines priorities, approves this BRD
Administrator	Maintains roles and permissions
All users	Their access is decided by the role they hold

2.2 User Roles

Role	Description	Main responsibilities	Access level
Administrator	Manages the whole system	View all roles; create, edit, duplicate and delete custom roles	Organization (full screen); always holds every permission
Manager	Assigns and monitors tasks	None on this screen	None by default (system role, read-only)
Employee	Works on assigned tasks	None on this screen	None by default (system role, read-only)
Custom role	Role defined on this screen	As granted	`roles.manage` opens the screen; `roles.view.all` shows all roles (otherwise only one's own); `roles.modify.access` allows changes (otherwise read-only)

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3. Scope

3.1 In Scope

- Listing roles with the number of users holding each, and searching them.
- Viewing a role's permissions as a tree or in a compare-roles matrix.
- Creating, duplicating, editing and deleting custom roles.
- Copying permissions from another role.
- Scope levels (for example Dashboard Own / Team / Organization) and permission prerequisites.
- Signing out holders of a role whose permissions change.

3.2 Out of Scope (this release)

- Giving a role to a user – Employee Management.
- Adding, changing or removing permissions in the catalogue – the catalogue is fixed by design.
- Changing the system roles Administrator, Manager and Employee.
- A history of role changes – not recorded.
- Permissions per individual user (outside a role) – not supported.

3.3 Modules Overview

#	Module	Purpose	Screens
1	Settings → Role Configuration	Roles and their permissions	Role Configuration

Change from original BRD

Original BRD item	Current requirement
Section 1.4: three fixed roles (Administrator, Manager, Employee)	The three roles remain as protected system roles; the organization can add custom roles with chosen permissions.
Screen 5 FR-5 "Assign appropriate roles"	Roles, including custom roles, are assigned in Employee Management; making someone an Administrator needs `roles.administrator.assign`.
Not in original BRD	Role Configuration is a new screen; access everywhere is decided by permissions, not by role names.

4. Business Process / Workflows

4.1 Create a Custom Role

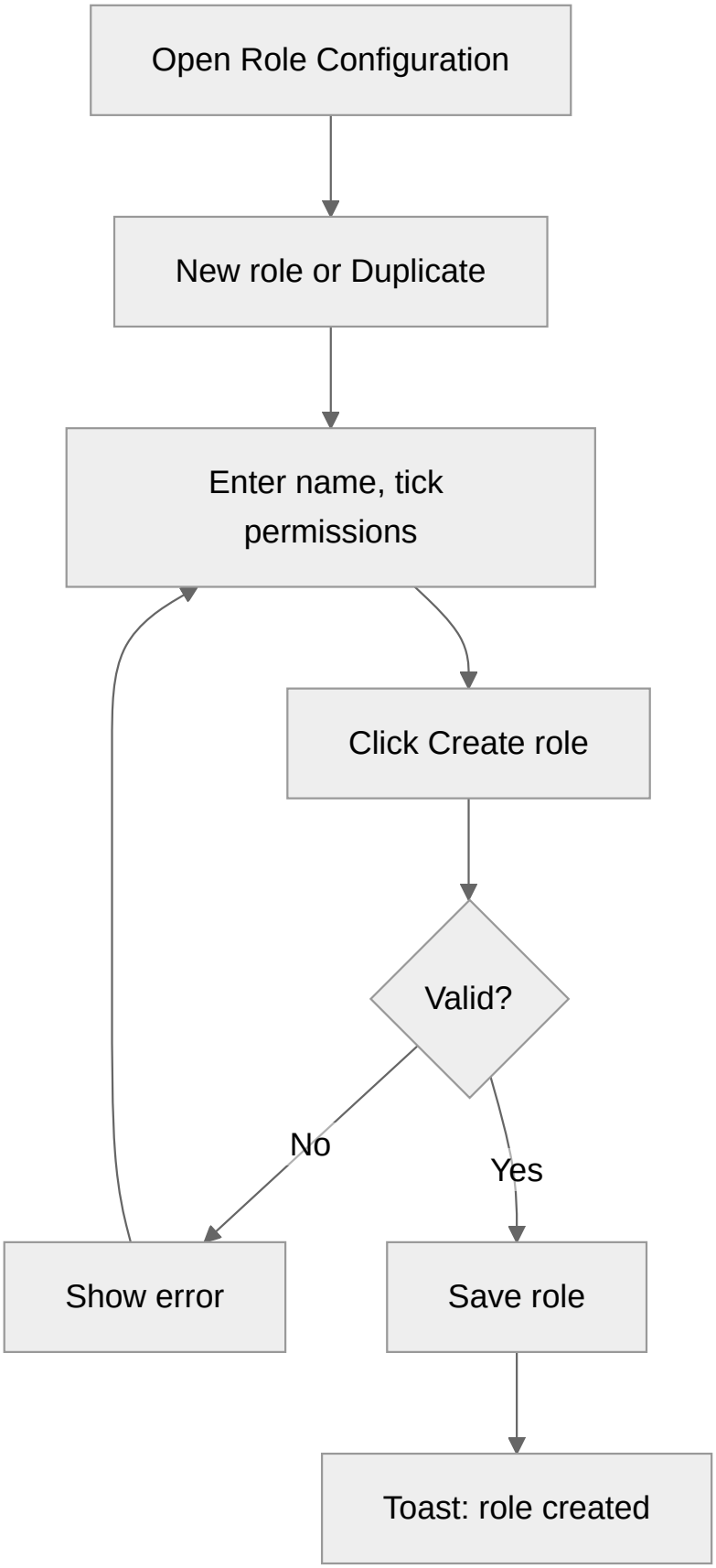
Trigger: A group needs access different from the system roles · **Actors:** Administrator, Custom role, System

Step	Actor / System	Action	Result
1	Administrator	Opens **Settings → Role Configuration**	Roles and the permission tree shown

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Step	Actor / System	Action	Result
2	Administrator	Clicks **New role**, or selects a role and clicks **Duplicate as custom role**	Empty draft, or a draft named "… copy" with that role's permissions
3	Administrator	Types the role name and ticks permissions; optionally **Copy permissions from…**	Prerequisites are ticked automatically; save bar counts the changes
4	Administrator	Clicks **Create role**	Name and permissions checked
5	System	Saves the role	Toast "`Role name` created"

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Example: The Administrator duplicates Manager, renames the draft "Team Lead", unticks Employee Management and keeps View all tasks, Create task and Workload. Creating it shows "Team Lead created". In Employee Management, Rahul is then given the role Team Lead.

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Exceptions / alternate flows:

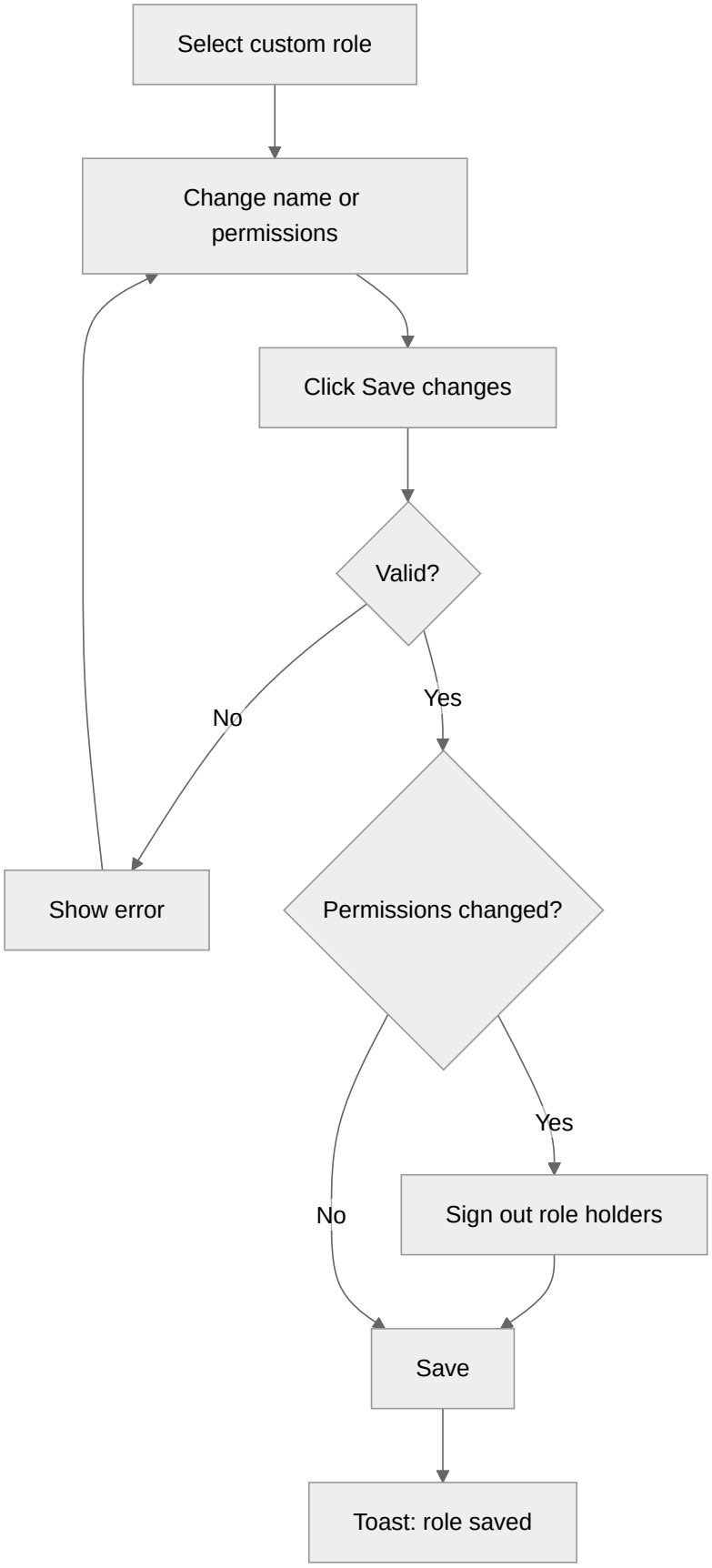
- Name blank: "Enter a name for the role."
- Name already used, in any letter case: "A role named "Team Lead" already exists."
- Permission without its prerequisite: "`permission` also needs `needed permission`."

4.2 Edit a Custom Role

Trigger: A custom role needs more or less access · **Actors:** Administrator, Custom role, System, role holders

Step	Actor / System	Action	Result
1	Administrator	Selects the custom role	Its permissions shown
2	Administrator	Renames it or ticks / unticks permissions	Save bar shows "N unsaved changes"
3	Administrator	Clicks **Save changes**	Checked as on create
4	System	If permissions changed, signs out every holder of the role	Holders sign in again with the new access
5	System	Saves the role	Toast "`Role name` saved"

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Example: The Administrator adds Reports to "Team Lead" and saves. Rahul, who holds Team Lead, is signed out; after signing in again he sees the Reports menu.

Exceptions / alternate flows:

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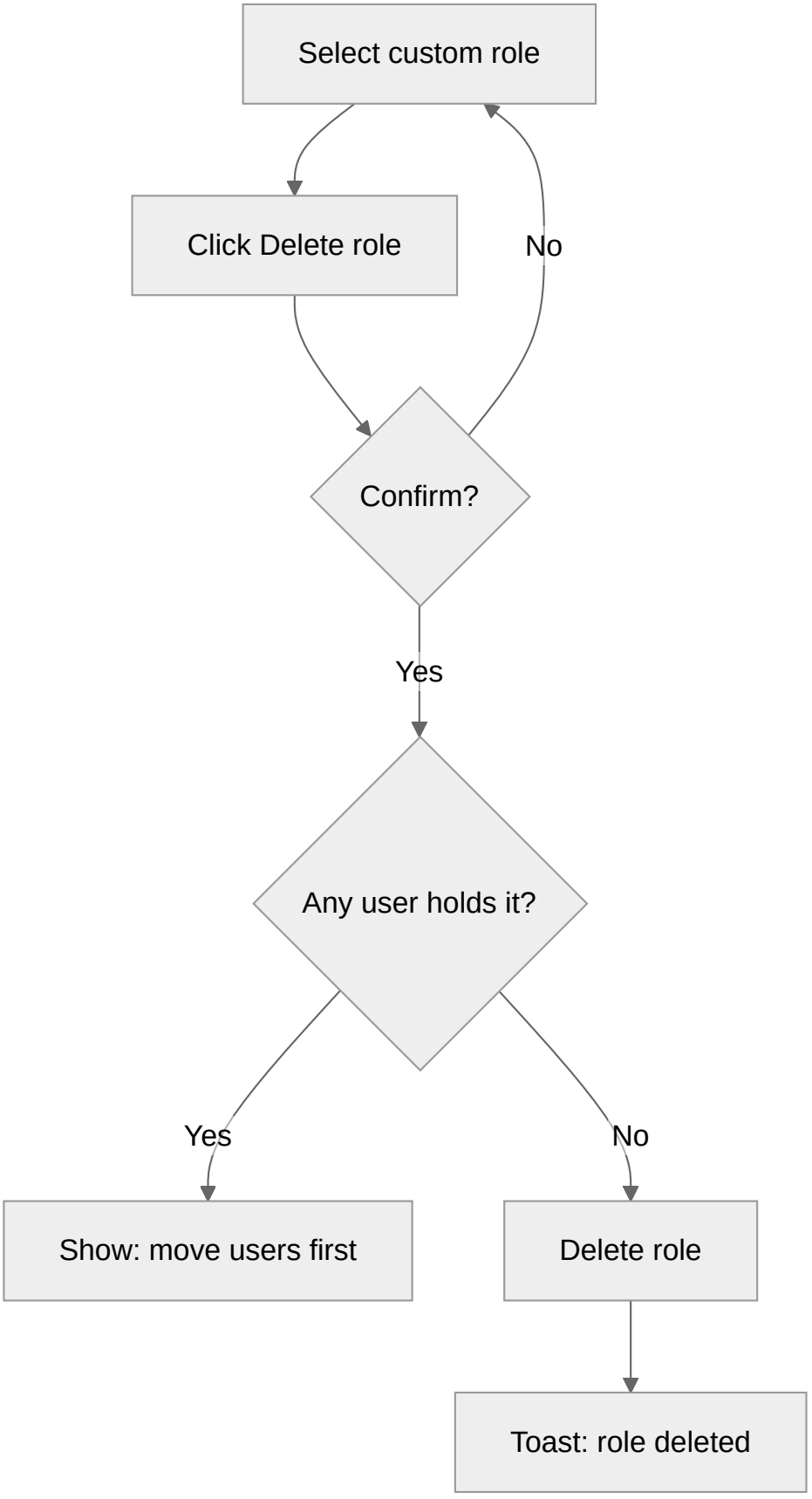
- Selecting another role with unsaved changes asks "Discard unsaved changes?" first.
- A name-only change does not sign anyone out.
- A system role cannot be saved: "`Role name` is a system role and is read-only: it cannot be changed or deleted."

4.3 Delete a Custom Role

Trigger: A custom role is no longer needed · **Actors:** Administrator, Custom role, System

Step	Actor / System	Action	Result
1	Administrator	Selects the custom role and clicks **Delete role**	Confirmation "Delete `Role name`?"
2	Administrator	Confirms	Request sent
3	System	Refuses when any user still holds the role	"2 users still hold this role. Move them to another role first."
4	System	Otherwise deletes the role and its permissions	Toast "`Role name` deleted"

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Example: "Team Lead" is held by Rahul and Priya. Deleting it is refused with "2 users still hold this role. Move them to another role first." After both are moved to Manager in Employee Management, the delete succeeds.

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Exceptions / alternate flows:

- The user cancels the confirmation: nothing changes.

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5. Screen-by-Screen Requirements

SCREEN 1: Role Configuration

Field	Value
Screen name	Role Configuration
Module	Settings
Access	Administrator; Custom role with `roles.manage` (+ `roles.view.all` to see all roles, + `roles.modify.access` to change). Manager and Employee: no access by default
Priority	Must

Purpose Choose exactly what each role can see and do, and manage custom roles.

Screen Elements

- Page title **Role Configuration**: "Choose exactly what each role can see and do. Administrator, Manager and Employee are protected; create your own roles for anyone else."
- **Roles** panel: **New role** button, **Find a role** search, list of roles (system roles first) with user counts.
- Role detail: role name box (custom) or name with **System role** badge and **Duplicate as custom role**; **Copy permissions from…**; **Delete role**; progress "N of M permissions", "x% enabled".
- **Search permissions** box; **All / Enabled / Disabled** filter; **Tree** / **Compare roles** view switch.
- Tree: module list, **Expand all / Collapse all**, **Select all / Deselect all**, tick boxes for module, screen and permission, scope selectors; tags **Always on** and **Administrator only**.
- Save bar: "N unsaved changes", **Discard**, **Create role** / **Save changes**.

Functional Requirements

ID	Requirement	Priority
REQ-ROLE-01	The system shall list all roles, system roles first, with the number of users holding each, and let the user search the list.	Must
REQ-ROLE-02	The system shall show only the user's own role when the user does not hold `roles.view.all`.	Must
REQ-ROLE-03	The system shall show a role's permissions as a tree grouped by module and screen.	Must
REQ-ROLE-04	The system shall show a compare view of every permission against every role.	Should
REQ-ROLE-05	The system shall let the user search permissions and filter them by All, Enabled or Disabled.	Should
REQ-ROLE-06	The system shall show system roles read-only and refuse any change or deletion of them.	Must

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ID	Requirement	Priority
REQ-ROLE-07	The system shall let the user create a custom role with a name and chosen permissions.	Must
REQ-ROLE-08	The system shall let the user duplicate any role as a new custom role named "... copy" with the same permissions.	Should
REQ-ROLE-09	The system shall let the user copy the permissions of another role into the role being edited.	Should
REQ-ROLE-10	The system shall require a role name of at most 100 characters that is unique regardless of letter case.	Must
REQ-ROLE-11	The system shall tick a permission's prerequisite automatically and refuse to save a permission without its prerequisite.	Must
REQ-ROLE-12	The system shall refuse to give `roles.administrator.assign` to any role other than Administrator.	Must
REQ-ROLE-13	The system shall keep always-on features (own profile, own dashboard, own tasks) on for every role, not switchable.	Must
REQ-ROLE-14	The system shall treat scope levels as inclusive, a higher level including the lower ones.	Must
REQ-ROLE-15	The system shall let the user rename a custom role and change its permissions.	Must
REQ-ROLE-16	The system shall count unsaved changes, let the user discard them, and ask "Discard unsaved changes?" before leaving a role with unsaved changes.	Should
REQ-ROLE-17	The system shall sign out every holder of a custom role when its permissions change, and not when only its name changes.	Must
REQ-ROLE-18	The system shall delete a custom role only after confirmation.	Must
REQ-ROLE-19	The system shall refuse to delete a role that any user still holds, stating how many users hold it.	Must
REQ-ROLE-20	The system shall keep the permission catalogue fixed and refuse any attempt to add, change or remove a permission.	Must
REQ-ROLE-21	The system shall give the Administrator role every permission again each time the application starts.	Must
REQ-ROLE-22	The system shall show everything read-only and hide **New role** when the user does not hold `roles.modify.access`.	Must
REQ-ROLE-23	The system shall show how many permissions of the catalogue a role has, as a count and a percentage.	Could
REQ-ROLE-24	The system shall hide the menu item from, and redirect to their dashboard, users without `roles.manage`.	Must
REQ-ROLE-25	The system shall confirm actions with "`Role name` created", "`Role name` saved" and "`Role name` deleted".	Should

Validations & Messages

Situation	Rule / message
Name blank	"Enter a name for the role."

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Situation	Rule / message
Name too long	"The role name can be at most 100 characters."
Duplicate name	"A role named "`name`" already exists."
Unknown permission	"Unknown permission: `keys`."
Missing prerequisite	""`permission`" also needs "`needed permission`"."
Administrator-only permission	"`permission` can only be held by the Administrator role and cannot be given to another role."
System role change	"`Role name` is a system role and is read-only: it cannot be changed or deleted."
Role in use	"1 user still holds this role. Move them to another role first." / "`n` users still hold this role. Move them to another role first."
Role not found	"That role does not exist."
Catalogue change attempt	"The permission list is fixed and cannot be changed."
Leaving with unsaved changes	"Discard unsaved changes?" – "The changes you made to this role have not been saved and will be lost."
Load failed	"Could not load roles and permissions."
Save / delete failed	"Could not save the role." / "Could not delete the role."
No permission	Menu item hidden; opening the address returns the user to their dashboard

> ⚠️ To be confirmed: `roles.manage` can be given to a custom role; a custom role holding `roles.manage` and `roles.modify.access` can create a role that may assign roles to users. Whether `roles.manage` should be Administrator-only. ⚠️ To be confirmed: a user with `roles.manage` but without `roles.view.all` sees only their own role in the list but can still open another role directly; whether that should be refused. ⚠️ To be confirmed: **Duplicate as custom role** is shown to users without `roles.modify.access`, but the draft cannot be saved; whether the button should be hidden. ⚠️ To be confirmed: a refusal for missing `roles.modify.access` shows only the general message "Could not save the role."; whether a specific message is wanted.

Expected Result The organization defines custom roles that grant exactly the screens and actions each group needs, while the system roles and the permission catalogue stay protected and changes reach the role's holders at once.

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6. Business Rules

ID	Rule	Applies to
BR-ROLE-01	Administrator, Manager and Employee are system roles: they cannot be renamed, changed or deleted.	SCREEN 1
BR-ROLE-02	The permission catalogue is fixed; nobody, including an Administrator, can change it.	SCREEN 1
BR-ROLE-03	Role names are unique, ignoring letter case ("team lead" equals "Team Lead"), and at most 100 characters.	SCREEN 1

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ID	Rule	Applies to
BR-ROLE-04	A permission that needs another can only be held together with it.	SCREEN 1
BR-ROLE-05	"Give the Administrator role" can only be held by the Administrator role.	SCREEN 1, Employee Management
BR-ROLE-06	Always-on features are granted to every user and are not stored as grants.	All modules
BR-ROLE-07	Scope levels include the lower ones: Organization includes Team and Own; View all tasks includes View my tasks.	All modules
BR-ROLE-08	A role that any user holds cannot be deleted.	SCREEN 1
BR-ROLE-09	A permission change on a custom role signs out all its holders.	SCREEN 1, Authentication
BR-ROLE-10	The Administrator role always holds every permission, including new ones.	All modules

Global rules that apply (see overall BRD section 6): only authorized users may create, change or manage records – authorization is decided by the permissions defined here; records with dependent data are not deleted in a way that damages history – a role in use cannot be deleted.

7. Non-Functional Requirements

ID	Category	Requirement	Measure / target
NFR-ROLE-01	Security	Access is checked on the server for every request, not only by hiding the menu	Requests without `roles.manage` / `roles.modify.access` are refused
NFR-ROLE-02	Security	New permissions apply without waiting for users to sign out	Holders of a changed role are signed out at once; grants are re-read on each request
NFR-ROLE-03	Integrity	System roles are protected at every level	Changes to system roles are refused, also by the stored data
NFR-ROLE-04	Usability	Large permission lists stay manageable	Search, filter, expand/collapse, select all and compare view available
NFR-ROLE-05	Usability	No accidental loss of edits	Unsaved-change counter and discard confirmation
NFR-ROLE-06	Performance	The screen loads and saves without noticeable delay	As the global performance NFR in the overall BRD
NFR-ROLE-07	Compatibility	Works in current desktop browsers, responsive layout	As the global compatibility NFR

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8. Data Requirements

Entity / data set	Purpose	Key fields	Retention / rules
Role	System and custom roles	Name, system flag, created at, number of holders	Name unique ignoring case; system roles never changed; a role in use cannot be deleted
Permission catalogue	The fixed list of permissions	Key, name, description, module, screen, section, action, prerequisite, scope level, assignable, always on	Fixed; maintained only with the application itself
Role permission grant	Which permissions a role holds	Role, permission	Removed together with the role
User sessions	Signed-in users of a role	User, session validity	Ended when the user's role permissions change

9. Integrations & External Dependencies

System / service	Purpose	Required?	Notes
Employee Management	Gives roles to users; users must be moved before a role is deleted	Required	Giving the Administrator role needs `roles.administrator.assign`
Authentication	Signs out holders of a changed role	Required	They sign in again with the new access
All modules	Every screen and action is controlled by the permissions defined here	Required	–

No external system is involved.

10. Reporting Requirements

Not applicable for this sub-module. The compare-roles view gives an on-screen overview of permissions per role.

11. Recommended Technology Stack (optional)

Not applicable for this sub-module. The technology stack is given in the overall BRD.

12. Assumptions, Constraints & Dependencies

Type	Description
Assumption	Each user holds exactly one role.

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Type	Description
Assumption	Only Administrators normally hold `roles.manage`.
Constraint	The permission catalogue is fixed; new permissions come only with a new application version.
Dependency	Employee Management assigns roles; Authentication ends sessions.

13. Risks

ID	Risk	Impact	Likelihood	Mitigation
R-ROLE-01	A custom role gets too much access	High	Medium	Compare view; prerequisites; Administrator-only permissions cannot be given away
R-ROLE-02	A custom role with role-management permissions widens its own or others' access	High	Low	Grant `roles.manage` only to trusted roles; see To be confirmed in SCREEN 1
R-ROLE-03	Holders lose unsaved work when signed out after a permission change	Medium	Low	Change roles outside busy hours; sign-out happens only when permissions change
R-ROLE-04	No history of role changes	Low	Medium	Compare view shows the current state

14. Acceptance Criteria

ID	Criterion	Verified by
AC-ROLE-01	The Administrator sees all roles, system roles first, with user counts, and can search them.	SCREEN 1, REQ-ROLE-01
AC-ROLE-02	A custom role with `roles.manage` but without `roles.view.all` sees only its own role.	REQ-ROLE-02
AC-ROLE-03	The tree and the compare view show a role's permissions; search and the Enabled filter narrow the list.	REQ-ROLE-03, REQ-ROLE-04, REQ-ROLE-05, REQ-ROLE-23
AC-ROLE-04	Manager cannot be renamed, changed or deleted; the read-only message is shown.	REQ-ROLE-06
AC-ROLE-05	"Team Lead" is created from scratch and by duplicating Manager; copying permissions from Employee fills the tree.	REQ-ROLE-07, REQ-ROLE-08, REQ-ROLE-09, REQ-ROLE-25
AC-ROLE-06	A blank name, a 101-character name and "team lead" when "Team Lead" exists are refused.	REQ-ROLE-10
AC-ROLE-07	Ticking "Edit employees" also ticks "View employees"; saving without the prerequisite is refused.	REQ-ROLE-11
AC-ROLE-08	Giving "Give the Administrator role" to a custom role is refused.	REQ-ROLE-12

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ID	Criterion	Verified by
AC-ROLE-09	Always-on features are shown as **Always on** and cannot be unticked; choosing Organization scope includes Team and Own.	REQ-ROLE-13, REQ-ROLE-14
AC-ROLE-10	Renaming and changing a custom role is saved; leaving with unsaved changes asks for confirmation.	REQ-ROLE-15, REQ-ROLE-16
AC-ROLE-11	After a permission change, holders are signed out; after a name-only change, they are not.	REQ-ROLE-17
AC-ROLE-12	Deleting a role held by 2 users is refused with the count; deleting an unused role after confirmation succeeds.	REQ-ROLE-18, REQ-ROLE-19
AC-ROLE-13	Any attempt to change the permission catalogue is refused with "The permission list is fixed and cannot be changed."	REQ-ROLE-20
AC-ROLE-14	After a restart, the Administrator role holds every permission in the catalogue.	REQ-ROLE-21
AC-ROLE-15	Without `roles.modify.access` the screen is read-only and **New role** is hidden; without `roles.manage` the menu item is hidden and the address redirects.	REQ-ROLE-22, REQ-ROLE-24

15. Requirements Traceability Matrix

Objective	Business rule(s)	Screen(s)	Functional requirement(s)	Acceptance criterion
OBJ-ROLE-01	BR-ROLE-03, BR-ROLE-04, BR-ROLE-06, BR-ROLE-07, BR-ROLE-08, global rules	SCREEN 1	REQ-ROLE-01, REQ-ROLE-03, REQ-ROLE-04, REQ-ROLE-05, REQ-ROLE-07, REQ-ROLE-08, REQ-ROLE-09, REQ-ROLE-10, REQ-ROLE-11, REQ-ROLE-13, REQ-ROLE-14, REQ-ROLE-15, REQ-ROLE-16, REQ-ROLE-18, REQ-ROLE-19, REQ-ROLE-23, REQ-ROLE-25	AC-ROLE-01, AC-ROLE-03, AC-ROLE-05, AC-ROLE-06, AC-ROLE-07, AC-ROLE-09, AC-ROLE-10, AC-ROLE-12
OBJ-ROLE-02	BR-ROLE-01, BR-ROLE-02, BR-ROLE-05, BR-ROLE-10, global rules	SCREEN 1	REQ-ROLE-02, REQ-ROLE-06, REQ-ROLE-12, REQ-ROLE-20, REQ-ROLE-21, REQ-ROLE-22, REQ-ROLE-24	AC-ROLE-02, AC-ROLE-04, AC-ROLE-08, AC-ROLE-13, AC-ROLE-14, AC-ROLE-15
OBJ-ROLE-03	BR-ROLE-09	SCREEN 1	REQ-ROLE-17	AC-ROLE-11

16. Project Deliverables

#	Deliverable	Format
1	Role Configuration screen with tree and compare views	Source code in the repository
2	System roles and the permission catalogue created at start	Initialization at application start
3	BRD, FRD, HLD and LLD for Role Configuration	Markdown
4	Tests for roles, permissions and the sign-out on change	Automated tests

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17. Glossary

Term	Meaning
System role	Administrator, Manager or Employee; built in and read-only
Custom role	A role created by the organization on this screen
Permission	One screen or action a role may use, for example `settings.reminders.manage`
Permission catalogue	The fixed list of all permissions
Prerequisite	A permission another permission needs (normally the screen's View)
Scope level	How far a permission reaches: Own, Team or Organization (or own / all tasks)
Always on	A feature every user has, such as their own profile
Holder	A user who has been given a role

18. Project Summary

Role Configuration turns TaskFlow's access control into something the organization can shape itself: the system roles stay protected, while custom roles are built by ticking permissions from a fixed catalogue, with prerequisites, scope levels and Administrator-only permissions enforced. Roles in use cannot be deleted, and permission changes sign out the holders so the new access applies at once.

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